Special Theory of Relativity

Unit 3: The Geometric Representation of Space-Time and Introduction to Cosmology

Vaibhav Yenare

Reference books

1. B. W. Carroll & D. A. Ostlie, *An Introduction to Modern Astrophysics*, Chs. 6, 17, 24, 27–29
2. J. V. Narlikar, *An Introduction to Relativity* (Cambridge Univ. Press, 2010)
3. T. Takeuchi, *An Illustrated Guide to Relativity* (Cambridge Univ. Press, 2010)
4. J. R. Taylor, *Classical Mechanics* (University Science Books, 2005), Ch. 15

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Part A The Geometric Representation of Space-Time

1 Where to find Part A

Syllabus topic (Unit 3)	Unit 1 notes, section
Space-time diagrams	§11 Space-time diagrams
Simultaneity	§10 Relativity of simultaneity; §15 on a space-time diagram
Length contraction and time dilation	§§12–13 (physics); §15 (geometry)
The interval, four-vectors, basic tensors	§17 Four-dimensional space-time, the interval s^2
The time order and space separation of events	§18 Light cone; timelike, spacelike, lightlike
The twin paradox	§19
The principle of equivalence and general relativity	§29
Gravitational red shift	§30

2 Summary of results

Geometry of space-time in one page.

1. **Space-time diagram:** x horizontal, ct vertical; events are points, histories are world lines; light rays are at 45°; material world lines are steeper than 45°.
2. **Moving frame:** the ct'-axis ($x' = 0$, i.e. $ct = x/\beta$) and the x'-axis ($t' = 0$, i.e. $ct = \beta x$) both tilt towards the light line by $\tan^{-1}\beta$.
3. **Simultaneity:** events simultaneous in S and separated by Δx are separated in S' by $\Delta t' = -\gamma V \Delta x/c^2$.
4. **Time dilation:** $\Delta t = \gamma \Delta\tau$; **length contraction:** $l = l_0/\gamma$.
5. **Interval:** $s^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 - c^2\Delta t^2$ is the same in all inertial frames.
6. **Classification:** $s^2 < 0$ timelike (order absolute; causal connection possible; proper time $\sqrt{-s^2}/c$); $s^2 = 0$ lightlike; $s^2 > 0$ spacelike (order frame-dependent; no causal connection; proper distance $\sqrt{s^2}$).
7. **Four-vectors:** $a'^\mu = \Lambda^\mu{}_\nu a^\nu$; $a \cdot b = -a^0 b^0 + \mathbf{a}\cdot\mathbf{b}$ invariant; $x^\mu = (ct, \mathbf{r})$, $u^\mu = \gamma(c, \mathbf{v})$, $p^\mu = (E/c, \mathbf{p})$, $k^\mu = (\omega/c, \mathbf{k})$.
8. **Tensors:** metric $g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$; $a_\mu = g_{\mu\nu}a^\nu$; $T'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta T^{\alpha\beta}$; tensor equations hold in all frames.
9. **Twin paradox:** the clock time along a world line is $\tau = \int \mathrm{d}t/\gamma$; the inertial (straight) world line between two events has the greatest proper time.
10. **Equivalence principle:** locally, gravity is indistinguishable from acceleration; freely falling frames are local inertial frames; gravity is curvature of space-time.

11. **Gravitational red shift:** $\Delta\nu/\nu = -gh/c^2 = -\Delta\Phi/c^2$; from a star, $z \approx GM/(Rc^2)$; clocks lower in a potential run slower.

Part B Introduction to Cosmology

3 The scale of the Universe

Cosmology is the study of the Universe as a whole: its contents, structure, origin and evolution. Its distances need new units:

Unit	Definition	Value
astronomical unit (AU)	mean Earth–Sun distance	$1.496 \times 10^{11}\,\mathrm{m}$
light-year (ly)	distance light travels in one year	$9.46 \times 10^{15}\,\mathrm{m}$
parsec (pc)	distance at which 1 AU subtends $1''$	$3.086 \times 10^{16}\,\mathrm{m} = 3.26\,\mathrm{ly}$
megaparsec (Mpc)	10^6 pc	$3.086 \times 10^{22}\,\mathrm{m}$

A ladder of structures:

- **Stars** are grouped into **galaxies** of 10^7–10^{12} stars. Our Milky Way is a barred spiral about 30 kpc across containing $\sim 10^{11}$ stars; the Sun is 8 kpc from its centre.
- Galaxies gather into **groups** (the Local Group: Milky Way, Andromeda M31, M33 and dozens of dwarfs, within $\sim$ 1 Mpc) and **clusters** (hundreds to thousands of galaxies, e.g. Virgo at 16.5 Mpc, Coma at 100 Mpc).
- Clusters form **superclusters**, filaments and walls surrounding huge empty **voids** – the cosmic web.
- The observable Universe extends to about 14 000 Mpc and contains $\sim 10^{11}$–10^{12} galaxies.

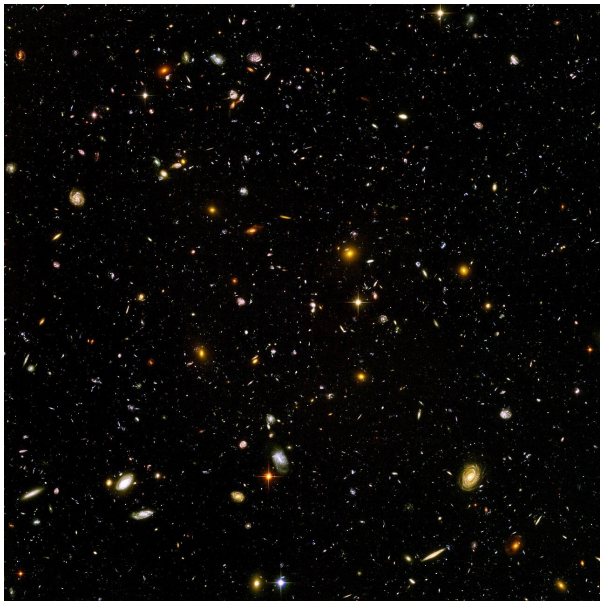


Figure 1: The Hubble Ultra Deep Field: almost every object in this tiny patch of sky (one tenth the width of the full Moon) is a galaxy; the faintest are seen as they were more than 12 billion years ago.
Source: Wikimedia Commons, "Hubble ultra deep field.jpg" (NASA and ESA; public domain)

4 Types of galaxies: the Hubble sequence

Until 1924 it was not known whether the "spiral nebulae" were inside our Galaxy. Edwin Hubble found Cepheid variable stars in the Andromeda nebula and showed that it lies far outside the Milky Way: the nebulae are galaxies in their own right. In 1926–1936 he classified them by appearance into the **tuning-fork diagram** (Fig. 2).

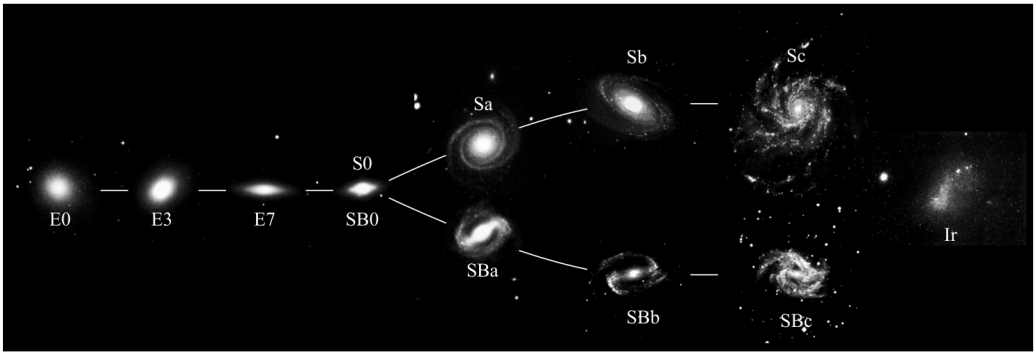


Figure 2: Hubble's tuning-fork diagram of galaxy types: ellipticals E0–E7 on the handle, the lenticular S0 at the junction, normal spirals Sa–Sc on the upper prong and barred spirals SBa–SBc on the lower prong.
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics*, Ch. 24, Fig. 1*

4.1 Elliptical galaxies (E)

Smooth, featureless ellipses of light with no spiral arms. They are classified **E***n* by their apparent flattening:

$$n = 10\Big(1 - \frac{b}{a}\Big),$$

where a and b are the apparent major and minor axes. E0 is circular, E7 the most flattened. Ellipticals contain mainly old, red stars, very little gas and dust and hence little star formation. They range from dwarf ellipticals ($10^7\,M_\odot$) to giant ellipticals such as M87 ($10^{13}\,M_\odot$, the largest galaxies known, at the centres of rich clusters). Stars move in randomly oriented orbits rather than in a rotating disc.

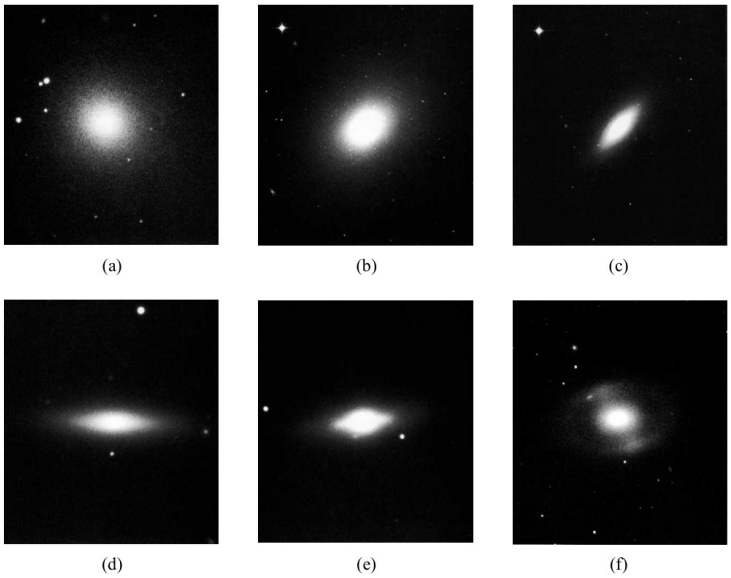


Figure 3: Typical early-type galaxies: (a) IC 4296 (E0), (b) NGC 4365 (E3), (c) NGC 4564 (E6), (d) NGC 4623 (E7), (e) NGC 4251 (S0), (f) NGC 4340 (barred S0).
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics*, Ch. 24, Fig. 4*

4.2 Lenticular galaxies (S0)

Transitional objects with a central bulge and a disc but *no* spiral arms and little gas – lens-shaped.

4.3 Spiral galaxies (S) and barred spirals (SB)

A flat rotating **disc** containing gas, dust, young blue stars and spiral arms, a central **bulge** of older stars, and a faint spherical **halo** with globular clusters. Along the sequence Sa → Sb → Sc:

- the bulge becomes smaller relative to the disc;
- the arms become more loosely wound and more clumpy;
- the fraction of gas and young stars increases.

In about half of all spirals the arms start from the ends of a straight **bar** through the nucleus: barred spirals SBa–SBc. The Milky Way is classified SBbc. Spiral arms are not fixed material structures (they would wind up); they are *density waves* in which star formation is triggered.

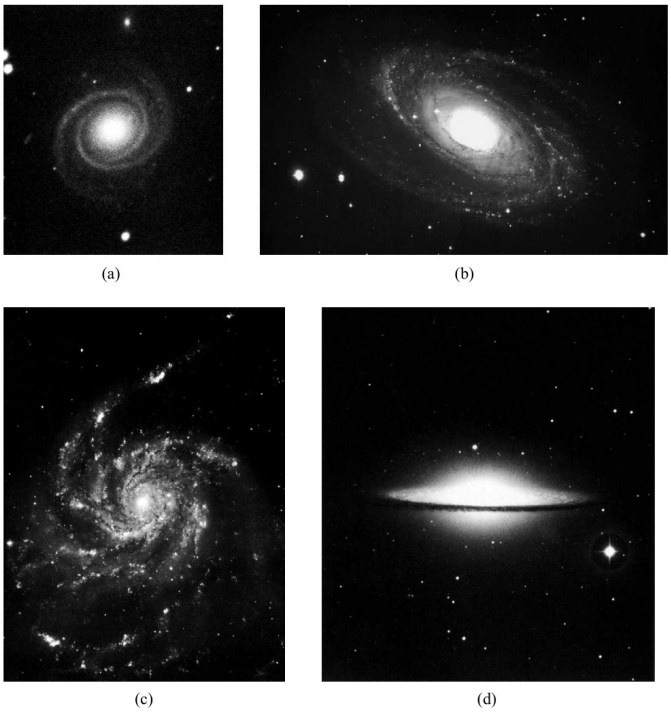


Figure 4: Normal spirals: (a) NGC 7096 (Sa), (b) M81 (Sb), (c) M101 (Sc), (d) M104, the Sombrero.
Source: Carroll & Ostlie, Ch. 24, Fig. 5

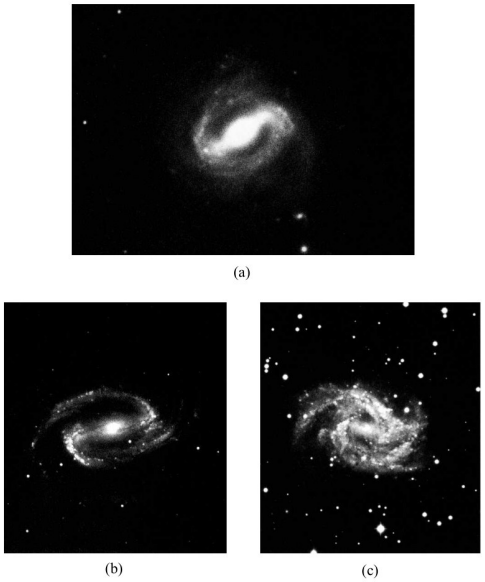


Figure 5: Barred spirals: (a) NGC 175 (SBab), (b) NGC 1300 (SBb), (c) NGC 2525 (SBc).
Source: Carroll & Ostlie, Ch. 24, Fig. 6



Figure 6: The Whirlpool galaxy M51, a "grand-design" spiral with two well-defined arms, interacting with a small companion.
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics, *Ch. 24, Fig. 17*

4.4 Irregular galaxies (Irr)

Galaxies with no regular shape or symmetry, usually small, rich in gas and young stars. Irr I show some structure (e.g. the Magellanic Clouds, satellites of the Milky Way); Irr II are chaotic, often disturbed by collisions (e.g. M82).

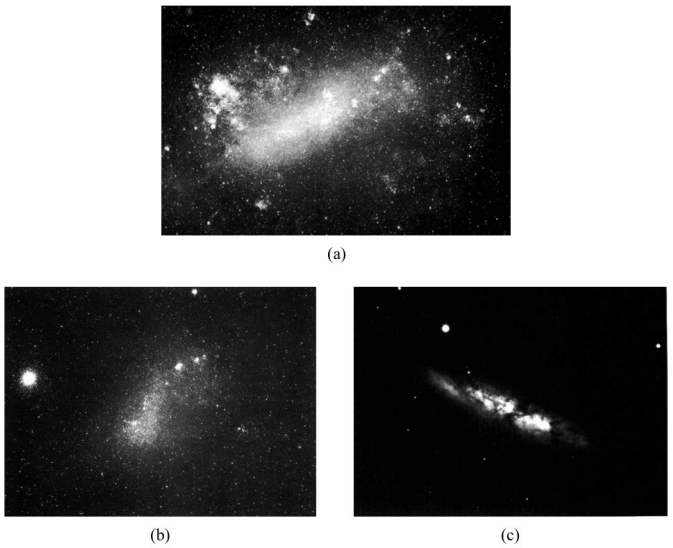


Figure 7: Irregular galaxies: (a) the Large Magellanic Cloud, (b) the Small Magellanic Cloud, (c) M82 (Irr II).
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics*, Ch. 24, Fig. 7*

Property	Elliptical	Spiral	Irregular
Shape	smooth ellipsoid	disc + bulge + arms	no symmetry
Stars	old, red	old in bulge/halo, young in arms	many young
Gas and dust	very little	plenty in disc	plenty
Star formation	almost none	ongoing in arms	vigorous
Motion of stars	random orbits	ordered rotation of disc	irregular
Mass ($M_\odot$)	10^7–10^{13}	10^9–10^{12}	10^8–10^{10}

"Early" and "late" types.

Hubble wrongly suspected that galaxies evolve from left to right along the tuning fork. The names *early-type* (E, S0) and *late-type* (Sc, Irr) survive, but they do not describe ages.

Dark matter.

The rotation speeds of stars and gas in the outer parts of spiral galaxies (measured from Doppler shifts, e.g. of the 21 cm line) stay nearly constant far beyond the visible disc ("flat rotation curves"). Most of a galaxy's mass must therefore be invisible *dark matter*.

Solved Example 1: Classifying an elliptical

An elliptical galaxy has apparent axes $a = 5.0'$ and $b = 3.5'$. Then $n = 10(1 - 3.5/5.0) = 3$: type E3. (The true shape may be flatter: a disc-like galaxy seen face-on looks E0.)

5 Radio sources, active galaxies and quasars

5.1 The discovery of cosmic radio sources

In 1932 Karl Jansky, investigating radio noise for Bell Telephone, found a steady hiss from the direction of the centre of the Milky Way – the birth of radio astronomy. Grote Reber mapped the radio sky with a home-made dish (1937–1944). After the Second World War, surveys (Cambridge

3C catalogue, Australian surveys) found hundreds of discrete radio sources. In 1954 Baade and Minkowski identified the strong source Cygnus A with a faint, distant galaxy: it emits about a million times more radio power than the Milky Way.

5.2 Radio galaxies

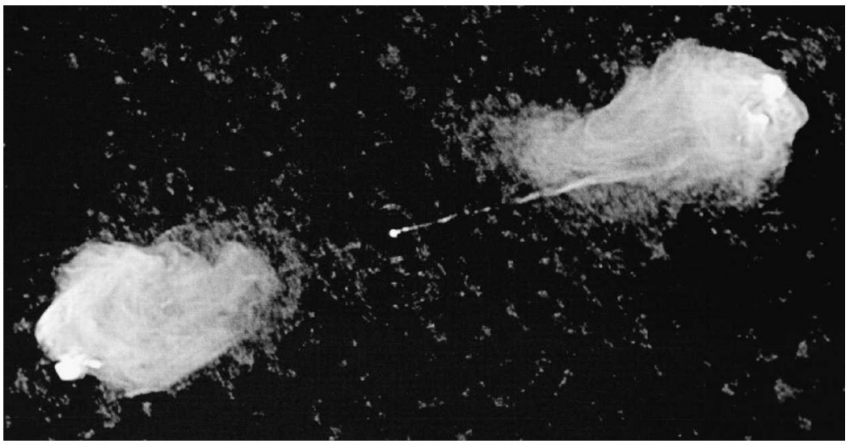


Figure 8: A VLA radio image of Cygnus A: two giant radio lobes about 100 kpc apart, fed by narrow jets from a compact core in the galaxy at the centre.
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics*, Ch. 28, Fig. 6*

A typical powerful radio galaxy has:

- a compact **core** at the nucleus of a (usually giant elliptical) galaxy;
- narrow **jets** of plasma moving outwards at relativistic speeds;
- two huge **lobes**, often far larger than the visible galaxy, where the jets are stopped by the intergalactic gas.

The radio emission is **synchrotron radiation** from relativistic electrons ($\gamma \gg 1$) spiralling in magnetic fields (Unit 2): it has a power-law spectrum (not a blackbody) and is strongly polarised. Because of relativistic beaming (Unit 1, aberration), a jet moving towards us looks much brighter than one moving away – many sources show only one jet.

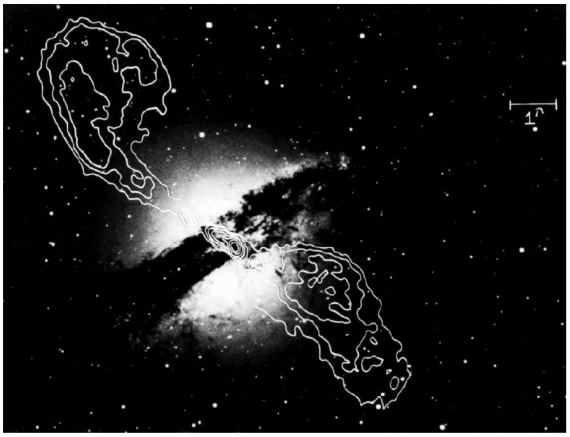


Figure 9: Centaurus A: optical image with radio contours of the inner lobes superimposed.
Source: Carroll & Ostlie, Ch. 28, Fig. 11

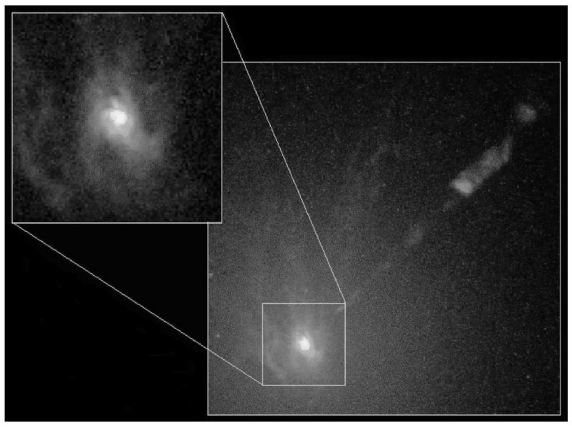


Figure 10: The giant elliptical M87 and its optical jet, emerging from the region around a supermassive black hole of $6.5 \times 10^9 M_\odot$.
Source: Carroll & Ostlie, Ch. 28, Fig. 10

5.3 Quasars

In 1960 the radio source 3C 48 was identified with a star-like object with a strange spectrum. In 1963 Maarten Schmidt realised that the emission lines of 3C 273 were simply the hydrogen Balmer lines shifted to the red by an unprecedented $z = 0.158$ (Fig. 12). Such "quasi-stellar radio sources" – **quasars** (QSOs) – are now known by the hundred thousand.

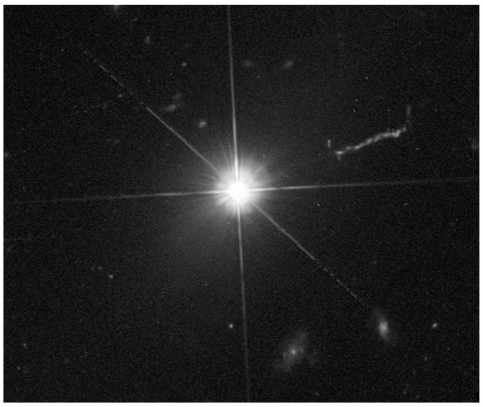


Figure 11: The quasar 3C 273 and its jet.
Source: Carroll & Ostlie, Ch. 28, Fig. 12

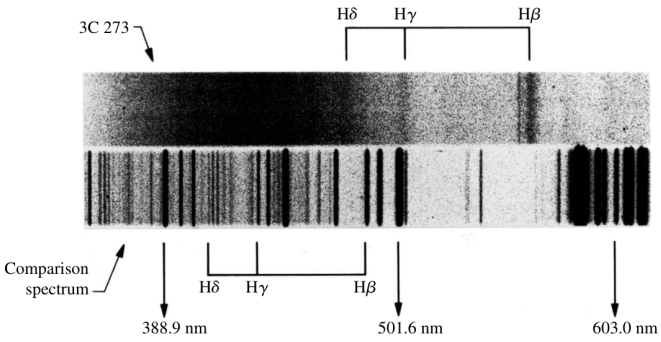


Figure 12: Spectrum of 3C 273 (top) above a laboratory comparison spectrum: the Balmer lines Hβ, Hγ, Hδ are shifted to longer wavelengths by $z = 0.158$.
Source: Carroll & Ostlie, Ch. 28, Fig. 13

Properties of quasars.

1. Star-like (point) images on ordinary photographs; deep images show a faint host galaxy.
2. Large red shifts ($z = 0.06$ to above 7): by Hubble's law they are very distant.
3. Enormous luminosity, 10^{38}–10^{41} W, up to 1000 times a whole galaxy – from a tiny region.
4. Rapid variability (days to months), which limits the size of the emitting region to light-days.

5. Broad emission lines (gas moving at thousands of km/s); a non-thermal continuum from radio to X-rays; only about 10% are strong radio sources.

The central engine. Quasars, radio galaxies and Seyfert galaxies (spirals with very bright, compact nuclei) are all **active galactic nuclei** (AGN). The accepted explanation: a *supermassive black hole* of 10^6–$10^{10} M_\odot$ accreting matter through a hot accretion disc, which converts up to $\sim 10\%$ of the rest energy mc^2 of infalling matter into radiation (compare 0.7% for nuclear fusion). In the *unified model* the different types of AGN are the same kind of object seen from different angles.

Solved Example 2: How big can a quasar be?

A quasar's brightness doubles in $\Delta t = 1\,\text{day}$. Different parts of a source cannot brighten in step faster than light can cross it, so its size is $R \lesssim c\Delta t = (3\times10^8)(86400) = 2.6\times10^{13}\,\text{m} \approx 170\,\text{AU}$ – about the size of our Solar System, yet it outshines a galaxy of 10^{11} stars.

5.4 Superluminal motion – relativity in the sky

Radio blobs in the jets of some quasars appear to move across the sky faster than light (Fig. 13). This is an illusion of light-travel time, not a violation of relativity.

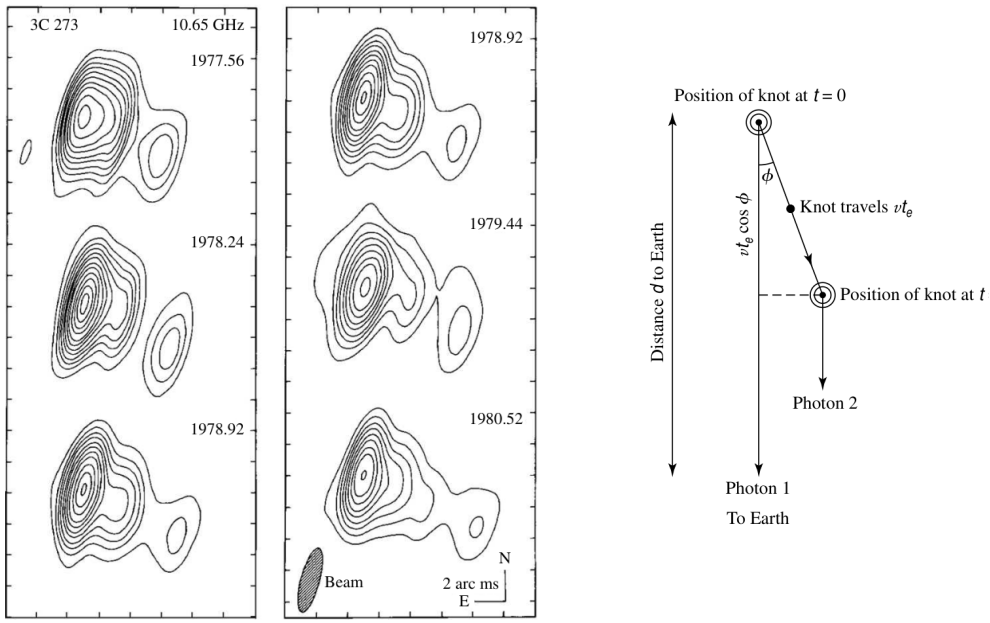


Figure 13: **Left:** a radio knot moving away from the core of 3C 273 (VLBI maps 1977–1980) with an apparent transverse speed of several times c. **Right:** geometry: the knot moves at speed v at an angle ϕ to our line of sight.

Source: Carroll & Ostlie, Ch. 28, Figs. 31 and 32

Derivation [5 marks]: Apparent transverse speed

A knot moves with speed v at angle ϕ to the line of sight. It emits one signal at $t = 0$ and a second at $t = t_e$. In that time it moves $vt_e\cos\phi$ towards us and $vt_e\sin\phi$ across the sky. The second signal has $vt_e\cos\phi$ less distance to travel, so the signals arrive separated by

$$\Delta t_{\text{obs}} = t_e - \frac{vt_e\cos\phi}{c} = t_e(1-\beta\cos\phi).$$

Apparent transverse speed:

$$v_{\rm app} = \frac{v t_e \sin\phi}{\Delta t_{\rm obs}} = \frac{v\sin\phi}{1-\beta\cos\phi}.$$

This is maximum when $\cos\phi = \beta$, giving $v_{\rm app,max} = \gamma v$, which exceeds c whenever $\beta > 1/\sqrt{2}$. ■

Solved Example 3: Superluminal knot

$\beta = 0.95$ ($\gamma = 3.20$). At $\phi = 18.2°$ ($\cos\phi = 0.95$): $v_{\rm app} = \gamma\beta c = 3.0c$. At $\phi = 40°$: $v_{\rm app} = \dfrac{0.95\sin 40°}{1-0.95\cos 40°}c = \dfrac{0.611}{0.272}c = 2.2c$.

6 Doppler shift and the expansion of the Universe

6.1 Red shifts of galaxies

Between 1912 and 1925 Vesto Slipher measured the Doppler shifts of about 40 spiral nebulae. Apart from a few nearby ones (Andromeda is approaching us at about 110 km/s), *almost all were red-shifted*: galaxies are moving away from us. The shift is measured from known spectral lines, e.g. the H and K lines of ionised calcium at 396.8 and 393.4 nm (Fig. 14).

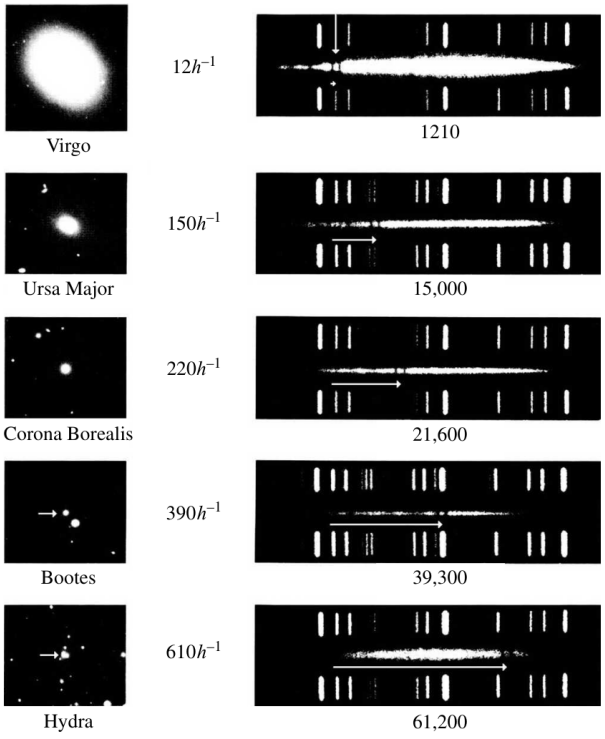


Figure 14: Galaxies in five clusters (Virgo, Ursa Major, Corona Borealis, Boötes, Hydra) and their spectra. The arrow shows the shift of the calcium H and K lines; the fainter (more distant) the galaxy, the larger the red shift. Recession velocities are given in km/s.
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics*, Ch. 27, Fig. 8*

Red shift.

$$z = \frac{\lambda_{\rm obs} - \lambda_{\rm rest}}{\lambda_{\rm rest}} = \frac{\Delta\lambda}{\lambda_{\rm rest}}.$$

- For $v \ll c$: $z \approx v/c$ (classical Doppler shift).
- Relativistic Doppler (Unit 1): $1 + z = \sqrt{\dfrac{1+\beta}{1-\beta}}$, i.e. $\beta = \dfrac{(1+z)^2 - 1}{(1+z)^2 + 1}$.
- **Cosmological red shift:** strictly, distant galaxies are not moving *through* space; space itself expands, and light waves are stretched with it. If $R(t)$ is the scale factor (the size of the Universe),

$$1 + z = \frac{R(t_{\rm obs})}{R(t_{\rm emit})} = \frac{\lambda_{\rm obs}}{\lambda_{\rm emit}}.$$

  A quasar at $z = 2$ emitted its light when the Universe was one third of its present size.

7 Hubble's law

In 1929 Edwin Hubble combined Slipher's velocities with his own distances (from Cepheid variables and bright stars) and found that the recession velocity is proportional to distance (Fig. 15). With Milton Humason he extended the law to much greater distances by 1931.

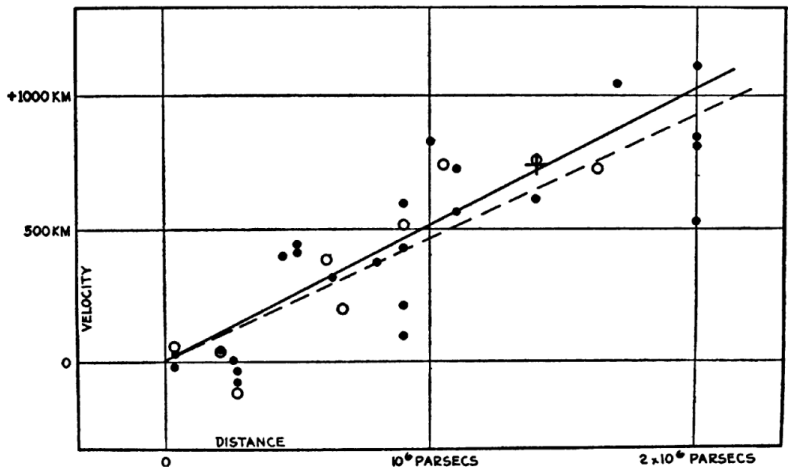


Figure 15: Hubble's original velocity–distance diagram (1929). The vertical units are actually km/s (not km). The slope – Hubble's original value of $H_0 \approx 500\,\mathrm{km/s/Mpc}$ – was far too large because his distances were underestimated.
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics, *Ch. 27, Fig. 7*

Hubble's law.

$$v = H_0\, d,$$

where H_0 is the **Hubble constant**: $H_0 \approx 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (current measurements give 67–73; the disagreement between methods is called the "Hubble tension"). Often written $H_0 = 100h\,\mathrm{km/s/Mpc}$ with $h \approx 0.7$.

- **Hubble time:** $t_H = 1/H_0 \approx 14\,\mathrm{Gyr}$ – the approximate age of the Universe.
- **Hubble distance:** $d_H = c/H_0 \approx 4300\,\mathrm{Mpc}$ – the scale of the observable Universe.



Figure 16: Einstein and Hubble at the Mount Wilson Observatory (1931). Einstein had built a static model of the Universe in 1917; after seeing the evidence for expansion he abandoned it.
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics, *Ch. 27, Fig. 9*

7.1 What Hubble's law means

Derivation [5 marks]: Hubble's law has no centre; the Hubble time

(a) No centre. Let galaxies A and B be at positions $\mathbf{r}_A$, $\mathbf{r}_B$ from us, with $\mathbf{v}_A = H_0\mathbf{r}_A$ and $\mathbf{v}_B = H_0\mathbf{r}_B$. An observer on A sees B at $\mathbf{r}_B - \mathbf{r}_A$ moving with velocity

$$\mathbf{v}_B - \mathbf{v}_A = H_0(\mathbf{r}_B - \mathbf{r}_A).$$

This is Hubble's law again, with the same H_0. Every observer in every galaxy sees all others receding according to the same law: we are not at the centre, and there is no centre. A linear velocity–distance law is the only one with this property (uniform expansion).

(b) Age. If each galaxy has moved with constant speed since all were together, a galaxy now at distance d took a time $t = d/v = d/(H_0 d) = 1/H_0$, the same for all galaxies. So $t_H = 1/H_0$ estimates the time since the expansion began (the Big Bang). For $H_0 = 70\,\mathrm{km/s/Mpc}$:

$$t_H = \frac{3.086 \times 10^{19}\,\mathrm{km}}{70\,\mathrm{km/s}} = 4.41 \times 10^{17}\,\mathrm{s} = 14.0\ \mathrm{Gyr}. \qquad \blacksquare$$

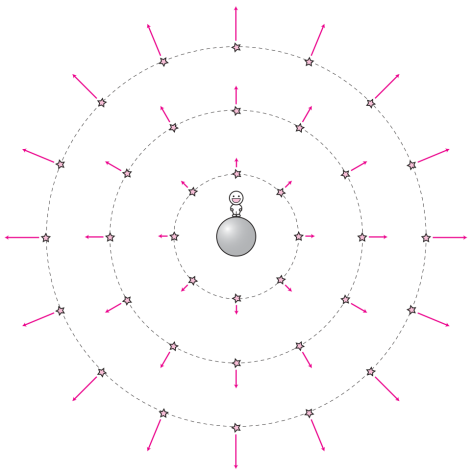


Figure 17: Uniform expansion: every galaxy moves away from every other, with speed proportional to distance. The picture looks the same from any galaxy – like raisins in a rising loaf, or dots on an inflating balloon.
Source: T. Takeuchi, An Illustrated Guide to Relativity, *§6.7.3*

7.2 Measuring distances: the distance ladder

Velocities are easy to measure (red shifts); distances are hard. Each method calibrates the next:

1. **Parallax** of nearby stars (up to a few kpc with the Gaia satellite).
2. **Cepheid variables**: their period is related to their luminosity (Henrietta Leavitt, 1912), so the period gives the luminosity, and comparison with the observed brightness gives the distance (up to $\sim 30\,\mathrm{Mpc}$ with HST).
3. **Tully–Fisher relation** for spirals (rotation speed $\leftrightarrow$ luminosity).
4. **Type Ia supernovae**: exploding white dwarfs of nearly standard peak luminosity, visible across billions of light-years. In 1998 they revealed that the expansion is *accelerating* ("dark energy").

Solved Example 4: Using Hubble's law

(a) A galaxy is 100 Mpc away. $v = H_0 d = 7000\,\mathrm{km/s}$, $z \approx v/c = 0.0233$. Its Hα line (656.3 nm) is observed at $656.3 \times 1.0233 = 671.6\,\mathrm{nm}$.

(b) The calcium K line (393.4 nm) of another galaxy is observed at 401.8 nm. $z = 8.4/393.4 = 0.0214$, $v \approx cz = 6400\,\mathrm{km/s}$, $d = v/H_0 = 91\,\mathrm{Mpc}$.

Solved Example 5: A distant quasar

A quasar has $z = 3$. The Universe was $1/(1+z) = 1/4$ of its present size when the light left it. (A naive Doppler interpretation would give $\beta = (16-1)/(16+1) = 0.88$; at such distances the cosmological interpretation must be used and "recession velocity" is not uniquely defined.)

8 The large-scale structure of the Universe

- **Groups and clusters.** The Local Group (~ 80 members within $\sim 1\,\mathrm{Mpc}$) is dominated by the Milky Way and Andromeda. Rich clusters such as Virgo and Coma contain thousands of galaxies, mostly ellipticals near the centre. The space between cluster galaxies is filled with hot gas at 10^7–$10^8\,\mathrm{K}$, which glows in X-rays (Fig. 34) and outweighs all the stars; the galaxies' speeds show that clusters are held together mainly by dark matter.
- **Superclusters** (e.g. the Local Supercluster, centred on Virgo; Laniakea, $\sim 160\,\mathrm{Mpc}$ across) are loose collections of clusters.
- **Redshift surveys** map galaxies in three dimensions (position on the sky + distance from red shift). The CfA survey (1986–1989) revealed the "Great Wall" and huge voids; the 2dF and Sloan surveys (2000s) mapped millions of galaxies (Fig. 19). Galaxies lie on thin **walls and filaments** around **voids** 20–100 Mpc across: the *cosmic web*.
- On scales larger than $\sim$ 100–300 Mpc the Universe looks the same everywhere and in every direction.

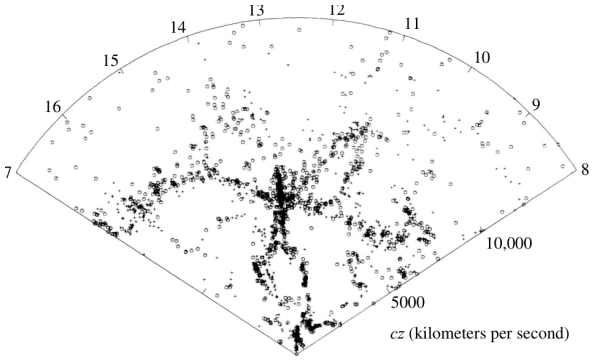


Figure 18: A slice of the CfA redshift survey: each dot is a galaxy; distance increases outwards with cz up to 15 000 km/s. Note the "stick-man" (Coma cluster), walls and voids.
Source: Carroll & Ostlie, Ch. 27, Fig. 24

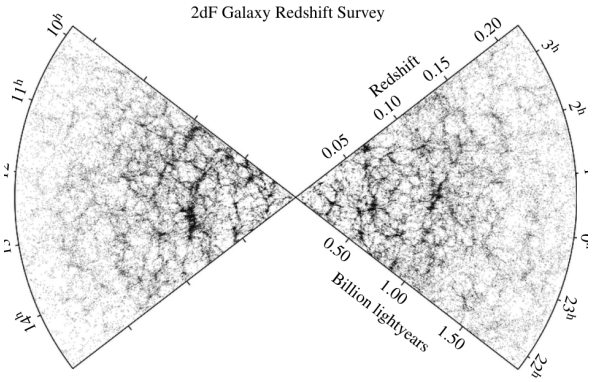


Figure 19: Two slices of the 2dF Galaxy Redshift Survey reaching $z \approx 0.25$: the cosmic web of filaments and voids.
Source: Carroll & Ostlie, Ch. 27, Fig. 26

The cosmological principle.
On large enough scales the Universe is **homogeneous** (the same at every place) and **isotropic** (the same in every direction). Hubble's law is the only expansion law consistent with it, and the uniformity of the cosmic microwave background (next section) is its strongest support.

9 The cosmic microwave background radiation

9.1 Prediction and discovery

If the Universe began hot and dense (the Big Bang), it was once filled with radiation in thermal equilibrium with matter. In 1948 Gamow, Alpher and Herman predicted that this radiation should survive today, cooled by the expansion to a few kelvin. In 1965 Arno Penzias and Robert Wilson at Bell Labs found an unexplained microwave noise in their horn antenna, equivalent to about 3 K, the same in every direction and at all times of day and year. R. H. Dicke and P. J. E. Peebles at Princeton immediately recognised it as the predicted radiation. (Nobel Prize 1978.)

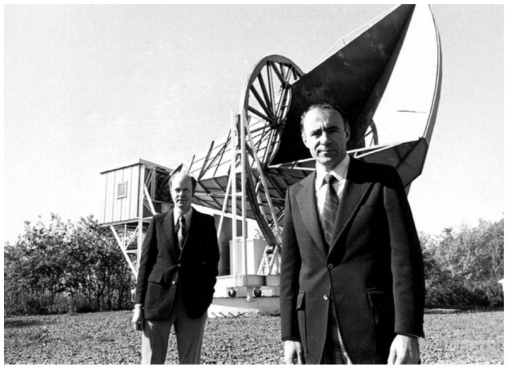


Figure 20: Robert Wilson and Arno Penzias in front of the horn antenna with which they discovered the cosmic microwave background in 1965.
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics, *Ch. 29, Fig. 8*

9.2 A perfect blackbody

The COBE satellite (1989–1993) measured the spectrum with exquisite precision: a perfect blackbody at $T_0 = 2.725\,\mathrm{K}$, with error bars smaller than the width of the plotted line (Fig. 21). (Nobel Prize 2006, Mather and Smoot.)

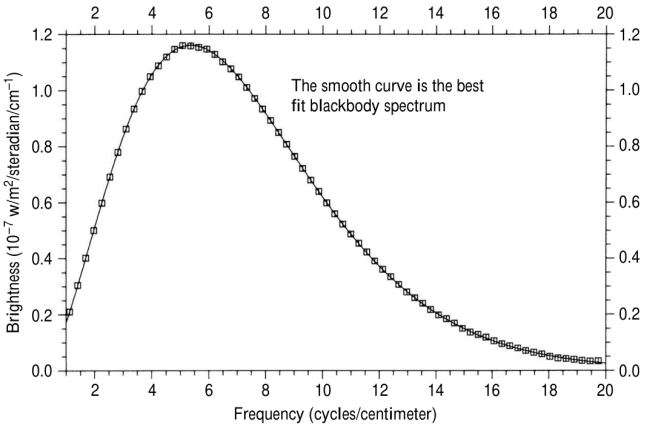


Figure 21: The COBE measurement of the spectrum of the cosmic microwave background (points) fitted by a blackbody curve of temperature 2.725 K.
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics, *Ch. 29, Fig. 9*

9.3 Origin

In the early Universe matter was ionised and photons were continually scattered by free electrons: the Universe was opaque. When the temperature fell to about 3000 K (some 380 000 years after the Big Bang, $z \approx 1100$), electrons and protons combined into hydrogen atoms (*recombination*) and the photons travelled freely ever after. Since then wavelengths have stretched by a factor $1 + z \approx 1100$, cooling the radiation from 3000 K to 2.7 K.

Derivation [5 marks]: The CMB temperature scales as $T \propto 1/R$, i.e. $T = T_0(1 + z)$
Every wavelength is stretched in proportion to the scale factor: $\lambda \propto R$. For a blackbody, Wien's law gives $\lambda_{\max} T = b$ (constant). If the radiation is a blackbody at temperature T_e when emitted, then after expansion its spectrum has the same shape with all wavelengths multiplied by $R_0/R_e = 1+z$, which is a blackbody of temperature

$$T_0 = \frac{b}{\lambda_{\max,0}} = \frac{b}{(1+z)\lambda_{\max,e}} = \frac{T_e}{1+z} \qquad \Rightarrow \qquad T(z) = T_0(1+z), \quad T \propto \frac{1}{R}.$$

(The energy density aT^4 then falls as R^{-4}: R^{-3} from dilution of photon number and R^{-1} from the red shift of each photon.) With $T_0 = 2.725\,\mathrm{K}$ and $z = 1100$: $T = 2.725 \times 1101 \approx 3000\,\mathrm{K}$. ■

9.4 Anisotropies

- **Dipole anisotropy** (Fig. 22): the CMB is slightly hotter in one direction and cooler in the opposite one, by $\Delta T/T \approx 1.2 \times 10^{-3}$. This is the Doppler effect of our own motion (Sun + Galaxy) at about 370 km/s relative to the CMB: $T(\theta) \approx T_0\big(1 + \frac{v}{c}\cos\theta\big)$. The CMB defines a "cosmic rest frame" – but this does *not* contradict relativity: it is a frame singled out by the matter and radiation in the Universe, not by the laws of physics.
- **Intrinsic fluctuations** of $\Delta T/T \sim 10^{-5}$ (COBE 1992; WMAP 2003–2010; Planck 2013–2018; Fig. 23). They are the imprint of tiny density variations at recombination – the seeds

from which gravity built galaxies and clusters. Their detailed pattern fixes the age (13.8 Gyr), the geometry (flat) and the composition of the Universe (about 5% ordinary matter, 27% dark matter, 68% dark energy).

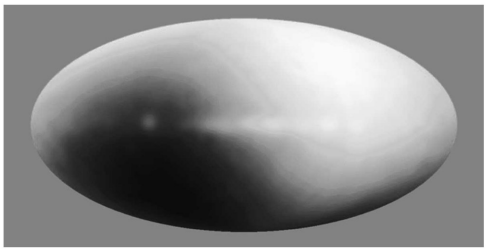


Figure 22: The CMB dipole anisotropy measured by COBE, caused by the Sun's motion relative to the CMB.
Source: Carroll & Ostlie, Ch. 29, Fig. 10

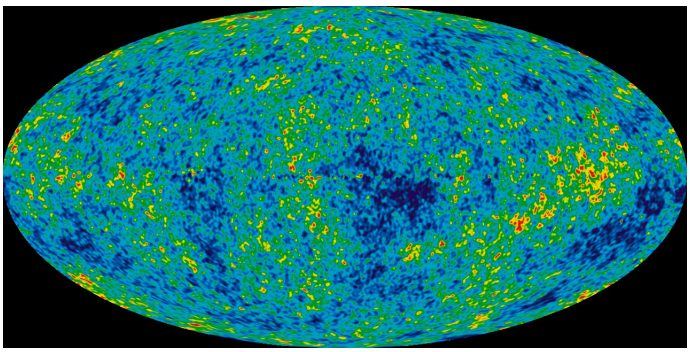


Figure 23: The whole-sky map of CMB temperature fluctuations ($\pm 200\,\mu$K) from nine years of WMAP data, after removing the dipole and the Milky Way.
Source: Wikimedia Commons, "Ilc 9yr moll4096.png" (NASA/WMAP Science Team; public domain)

Evidence for the hot Big Bang.
(1) Hubble's law – the expansion of the Universe; (2) the cosmic microwave background, a perfect blackbody of 2.725 K; (3) the abundances of the light elements (about 24% helium by mass, traces of deuterium and lithium) formed in the first few minutes; (4) evolution: distant (young) galaxies and quasars look different from nearby ones. The rival Steady State theory (Bondi, Gold, Hoyle, 1948) could not explain the CMB. (Narlikar, who worked with Hoyle, discusses alternative cosmologies in his Chapter 18.)

Solved Example 6: Numbers for the CMB

(a) Peak wavelength: $\lambda_{\max} = b/T = 2.898 \times 10^{-3}/2.725 = 1.06 \times 10^{-3}\,\mathrm{m} = 1.06\,\mathrm{mm}$ (microwaves).

(b) Energy density: $u = aT^4 = (7.566 \times 10^{-16})(2.725)^4 = 4.2 \times 10^{-14}\,\mathrm{J\,m^{-3}}$; there are about 410 CMB photons in every cm^3.

(c) Dipole: $\Delta T = T_0 v/c = 2.725 \times (370/3 \times 10^5) = 3.4\,\mathrm{mK}$.

10 Astronomy in different bands of radiation

Astronomical objects emit across the whole electromagnetic spectrum, and each band reveals different physics. By Wien's law ($\lambda_{\max} T = 2.9 \times 10^{-3}\,\mathrm{m\,K}$) cold dust and gas ($\sim$ 10–100 K) shine in the infrared and radio, stars (3000–30 000 K) in the optical and ultraviolet, and gas at millions of kelvin in X-rays. Non-thermal processes (synchrotron radiation, 21 cm line) dominate in the radio.

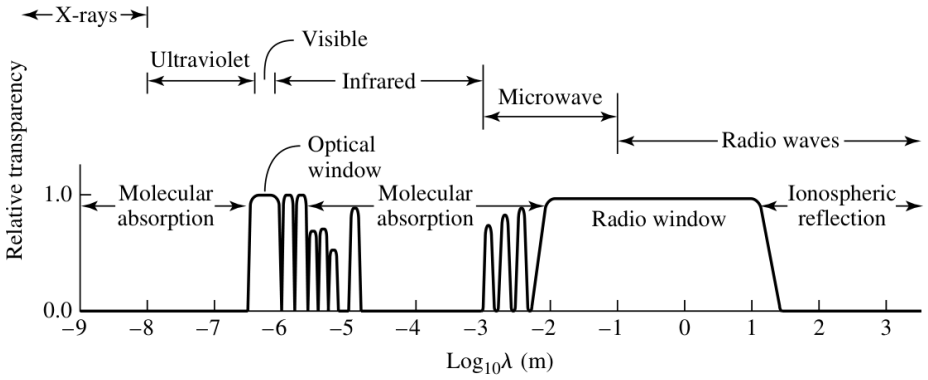


Figure 24: The transparency of the Earth's atmosphere as a function of wavelength. Only the optical window and the radio window (about 1 cm to 10 m) reach the ground; X-rays, ultraviolet and most of the infrared are absorbed, so these bands need rockets, balloons or satellites.
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics*, Ch. 6, Fig. 25*

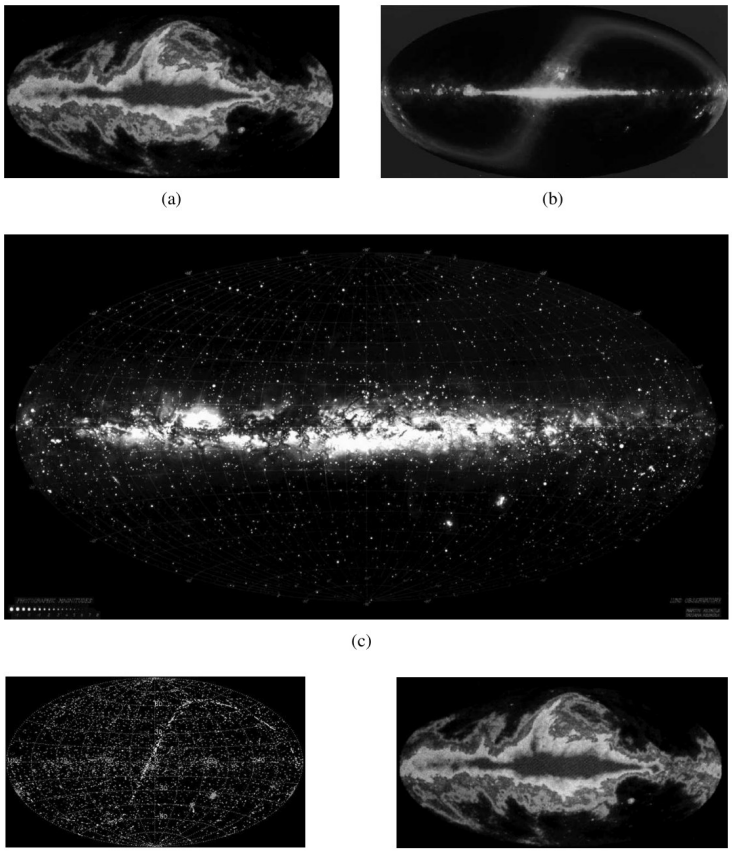


Figure 25: The whole sky in different wavebands; the plane of the Milky Way runs across the middle of each map: (a) radio, (b) infrared, (c) visible, (d) ultraviolet, (e) γ-ray. Each band shows a different Galaxy.
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics*, Ch. 6, Fig. 28*

10.1 Telescope basics

- **Light-gathering power** $\propto$ area $\propto D^2$ (D = aperture diameter).
- **Resolving power** (Rayleigh criterion): smallest resolvable angle $\theta_{\min} = 1.22\,\lambda/D$ (radians).

- **Magnification** = $f_{\text{objective}}/f_{\text{eyepiece}}$ (least important).

10.2 Optical astronomy

- Window 400–700 nm. Refractors (lenses) are limited to about 1 m (Yerkes 40-inch, 1897) by sagging and chromatic aberration; all large telescopes are reflectors: Keck (10 m), Gemini, Subaru; the 39 m Extremely Large Telescope is under construction. In India: the 3.6 m Devasthal telescope (ARIES, Nainital) and the 2.3 m Vainu Bappu telescope (Kavalur).
- On the ground, atmospheric turbulence ("seeing") blurs images to ~ 0.5–$1''$; observatories are built on high, dry mountains (Fig. 26); *adaptive optics* corrects the blurring in real time. In space the Hubble Space Telescope (1990, 2.4 m) reaches its diffraction limit of $0.05''$.
- Detectors: photographic plates, now CCDs; spectrographs give red shifts, compositions and temperatures.
- Optical astronomy gave us the Hubble sequence, red shifts and Hubble's law, quasar spectra and Type Ia supernova distances.



Figure 26: The observatories on Mauna Kea, Hawaii (4200 m), above much of the atmosphere's water vapour.
Source: Carroll & Ostlie, Ch. 6, Fig. 11

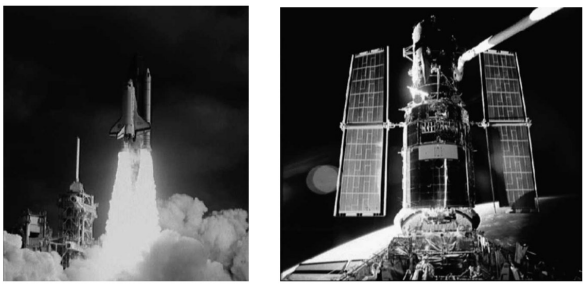


Figure 27: (a) Launch of the Hubble Space Telescope (1990); (b) HST during a servicing mission.
Source: Carroll & Ostlie, Ch. 6, Fig. 12

10.3 Radio astronomy

- Window from about 1 cm to 10 m; observations possible day and night and through clouds.
- Radio waves are long, so a single dish has poor resolution: a 100 m dish at 21 cm resolves only $\sim 9'$. Giant dishes: Arecibo (305 m, 1963–2020), FAST in China (500 m, 2016).
- **Interferometry**: signals from two antennas a distance d apart are combined. A source at angle θ reaches one antenna later by a path difference $d\sin\theta$, producing interference fringes; the resolution is $\theta \sim \lambda/d$, as if the telescope were d wide (Fig. 29). **Aperture synthesis** uses many antennas and the Earth's rotation to build up images: the Very Large Array (27 dishes over 36 km), and Very Long Baseline Interferometry across continents (resolution $< 0.001''$; the Event Horizon Telescope imaged the shadow of the M87 black hole in 2019).
- India: the **Giant Metrewave Radio Telescope** (GMRT, Khodad near Pune; NCRA–TIFR): 30 dishes of 45 m spread over 25 km, observing at 150–1450 MHz; and the Ooty Radio Telescope.

- **Discoveries:** radio galaxies and quasars; the 21 cm line of neutral hydrogen (1951), which maps the spiral arms and rotation of the Galaxy; pulsars (1967); the cosmic microwave background (1965); molecules in interstellar clouds.

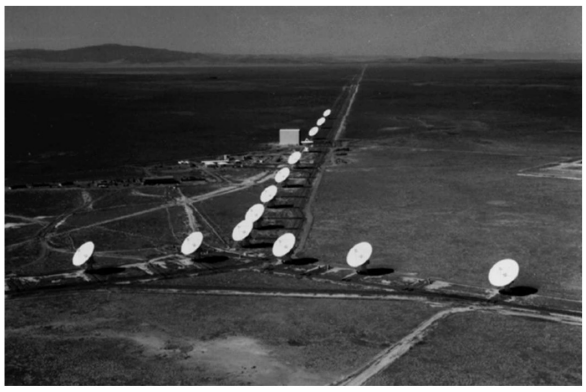


Figure 28: The Very Large Array near Socorro, New Mexico.
Source: Carroll & Ostlie, Ch. 6, Fig. 24

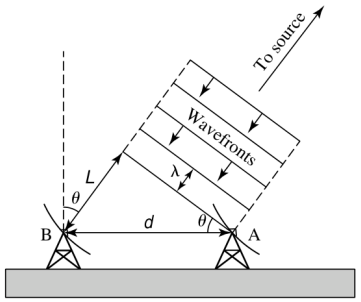


Figure 29: Radio interferometry: path difference $L = d \sin\theta$ between antennas A and B.
Source: Carroll & Ostlie, Ch. 6, Fig. 22

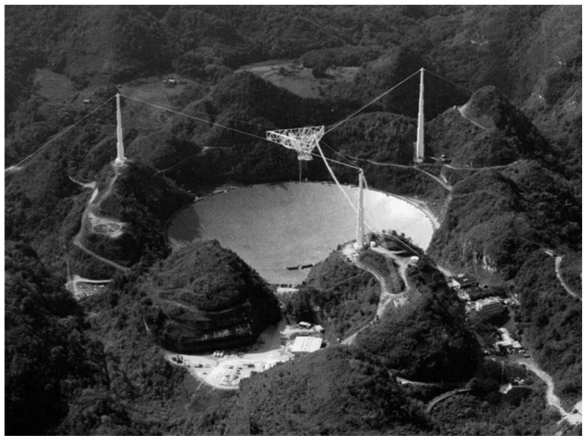


Figure 30: The 305 m Arecibo radio telescope, Puerto Rico.
Source: Carroll & Ostlie, Ch. 6, Fig. 21



Figure 31: An antenna of the Giant Metrewave Radio Telescope (GMRT) near Pune.
Source: Wikimedia Commons, "GMRT antenna at sunset.jpg" (CC BY 2.0)

10.4 X-ray astronomy

- X-rays (0.01–10 nm, 0.1–100 keV) are absorbed high in the atmosphere, so X-ray astronomy began only with rockets: in 1962 Riccardo Giacconi's team discovered the first cosmic X-ray source, Sco X-1 (Nobel Prize 2002). The satellite Uhuru (1970) found hundreds of sources.

- X-rays pass through (or are absorbed by) ordinary mirrors; they reflect only at *grazing incidence* (angles below $\sim 1^\circ$). X-ray telescopes therefore use nested, nearly cylindrical (Wolter) mirrors. Imaging missions: Einstein Observatory (1978), ROSAT (1990), **Chandra** (NASA, 1999, resolution $0.5''$), **XMM-Newton** (ESA, 1999), and India's **AstroSat** (ISRO, 2015).

- **Sources:** gas at 10^6–10^8 K – the solar corona, supernova remnants (Fig. 33), hot gas in galaxy clusters (Fig. 34); accretion discs around neutron stars and black holes in X-ray binaries (Cygnus X-1 was the first black-hole candidate); active galactic nuclei.

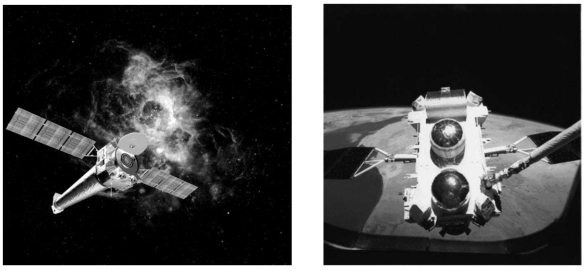


Figure 32: (a) Artist's view of the Chandra X-ray Observatory; (b) the Compton Gamma Ray Observatory being deployed.
Source: Carroll & Ostlie, Ch. 6, Fig. 27

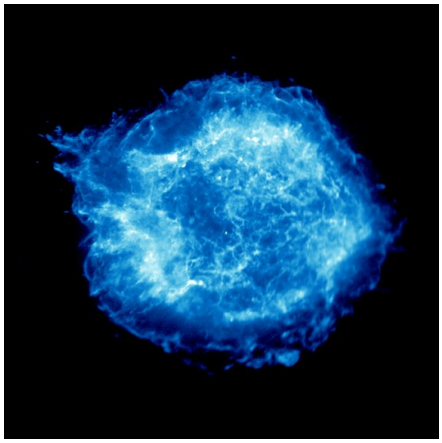


Figure 33: Chandra X-ray image of the supernova remnant Cassiopeia A, gas heated to millions of kelvin.
Source: Wikimedia Commons, NASA/CXC/GSFC/U. Hwang et al. (public domain)

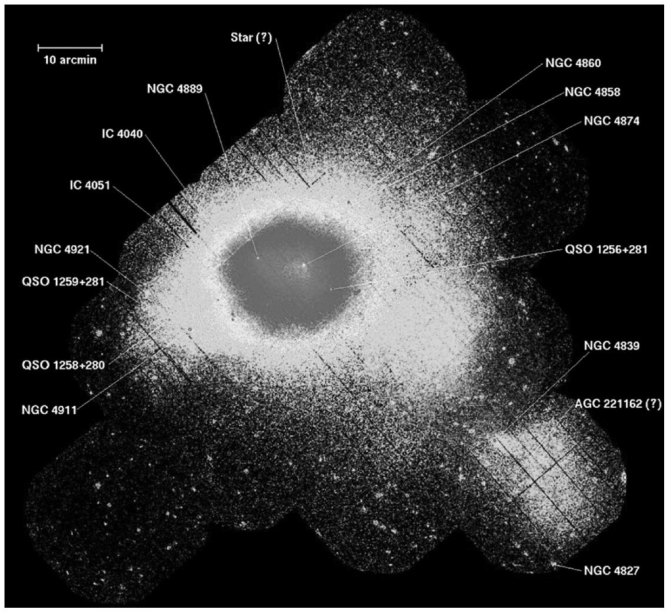


Figure 34: An XMM-Newton X-ray image of the Coma cluster. The diffuse glow is intracluster gas at about 10^8 K, which contains more mass than all the cluster's galaxies.
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics*, Ch. 27, Fig. 16*

10.5 Other bands (for completeness)

Infrared (Spitzer, James Webb Space Telescope 2021): cool stars, dust, star-forming regions, very red-shifted early galaxies. **Ultraviolet**: hot young stars, quasars. **Gamma rays** (Compton GRO,

Fermi): pulsars, gamma-ray bursts, blazars.

Solved Example 7: Resolving power

(a) A 10 m optical telescope at $\lambda = 550\,\text{nm}$: $\theta_{\min} = 1.22(550 \times 10^{-9})/10 = 6.7 \times 10^{-8}\,\text{rad} = 0.014''$ (in practice limited by seeing to $\sim 0.5''$ without adaptive optics).

(b) A 100 m radio dish at $\lambda = 21\,\text{cm}$: $\theta_{\min} = 1.22(0.21)/100 = 2.6 \times 10^{-3}\,\text{rad} = 8.8'$.

(c) GMRT with a 25 km baseline at 21 cm: $\theta \sim \lambda/d = 0.21/(2.5 \times 10^4) = 8.4 \times 10^{-6}\,\text{rad} = 1.7''$.

Solved Example 8: Why hot gas glows in X-rays

Intracluster gas at $T = 10^8\,\text{K}$: $\lambda_{\max} = 2.9 \times 10^{-3}/10^8 = 2.9 \times 10^{-11}\,\text{m} = 0.029\,\text{nm}$, and the typical photon energy $kT = (8.62 \times 10^{-5}\,\text{eV/K})(10^8\,\text{K}) = 8.6\,\text{keV}$ – hard X-rays. The solar corona at $2 \times 10^6\,\text{K}$ peaks near 1.5 nm (soft X-rays).

Summary of Unit 3, Part B

Key facts and formulae.

Hubble sequence	E0–E7 ($n = 10(1 - b/a)$), S0, Sa–Sc, SBa–SBc, Irr
Radio galaxies	core, relativistic jets, lobes; synchrotron radiation
Quasars	star-like, high z, $L \sim 10^{38}$–$10^{41}\,\text{W}$, small (variability), supermassive black holes
Superluminal motion	$v_{\text{app}} = v \sin\phi/(1 - \beta\cos\phi)$, max γv
Red shift	$z = \Delta\lambda/\lambda$; $z \approx v/c$; $1 + z = \sqrt{(1+\beta)/(1-\beta)}$; cosmological $1 + z = R_0/R_e$
Hubble's law	$v = H_0 d$, $H_0 \approx 70\,\text{km/s/Mpc}$, $t_H = 1/H_0 \approx 14\,\text{Gyr}$
Large-scale structure	groups, clusters, superclusters, filaments, voids; cosmological principle
CMB	blackbody 2.725 K, $\lambda_{\max} = 1.06\,\text{mm}$; $T = T_0(1 + z)$; dipole from our 370 km/s motion; $\Delta T/T \sim 10^{-5}$
Telescopes	light grasp $\propto D^2$; $\theta_{\min} = 1.22\lambda/D$; interferometer $\theta \sim \lambda/d$
Bands	optical (ground + HST), radio (window 1 cm–10 m; interferometry; GMRT), X-ray (space only; grazing-incidence mirrors; Chandra, XMM, AstroSat)